

Sunkanghong Wang (Last updated: 2026-05-11)

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I. EDUCATION

- Guangdong University of Technology** [🏢] September 2023 - June 2026 (expected)
Master of Engineering in Mechanical Engineering (GPA: 92.4/100 or 4.24/5 or 3.68/4) Guangzhou, China
◦ Ranked 3rd and 1st out of 514 students (Top 1%) in the comprehensive grade-level assessments of 2024 and 2025
- Guangdong University of Technology** [🏢] ARWU(2025): 301-400 September 2019 - June 2023
Bachelor of Engineering in Industrial Engineering (GPA: 88.8/100 or 3.88/5 or 3.37/4) Guangzhou, China
◦ Ranked 4th out of 85 (Top 5%) for GPA and 6th out of 743 (Top 1%) for comprehensive score, recommended for post-graduate study

II. AWARDS & HONORS

Competitions: Won 10 national-level, 8 provincial-level, and 1 school-level awards.
Scholarships: Awarded 3 national (Top 0.5%), 5 first-class, and 1 corporate (Top 1%) scholarships.
Patents: Awarded 11 invention patents, 9 of which as the first (student) author, including 1 US patent.
Others: Alibaba cloud expert blogger; 24k+ followers, 5.8M+ reads, 18k+ favorites, and 84k+ shares on CSDN.
Selected Awards and Honors are listed below. Further details can be found on MyWebSite.

- National Scholarships (Top 0.5%)** September 2023, 2024, 2025
Awarding Body: Ministry of Education of the People's Republic of China (Scholarship)
- MeGa Outstanding Student Scholarship (Top 1%)** May 2025
Awarding Body: MeGa Precision Metal Technology (Dongguan) Co., Ltd (Scholarship)
- Special Prize of the Guangdong Mechanical Engineering Society Science and Technology Award** March 2025
Awarding Body: Guangdong Mechanical Engineering Society (Competition)
- Second Prize of "HuaWei Cup" China Post-Graduate Mathematical Contest in Modeling** December 2023
Awarding Body: Chinese Society for Academic Degrees and Postgraduate Education (Competition)
- First Prize of the Elite Challenge of Integrated Circuit EDA Design** December 2022
Awarding Body: Chinese Institute of Electronics (Competition)
- Dynamic Optimization Method and System for Procurement Specifications of BEF Raw Materials** July 2023
Awarding Body: United States Patent and Trademark Office (US Patent)
- Student Representative of the National Scholarship for Graduate Students (Top $\approx$ 0.003%)** May 2026
Awarding Body: Ministry of Education of China (Honor)
- Top Ten Outstanding Graduates of Guangdong University of Technology** June 2023
Awarding Body: Guangdong University of Technology (Honor)

III. SELECTED PUBLICATIONS

C=CONFERENCE, J=JOURNAL, R=UNDER REVIEW, $\bar{R}$ ($\bar{R}$)=MAJOR(MINOR) REVISION, P=IN PROGRESS

- [J.1] (JCR Q1; ABS 4) Wang, S., Baldacci, R., Liu, Q., & Wei, L. (2025). EATKG: An open-source efficient exact algorithm for the two-dimensional knapsack problem with guillotine constraints. *European Journal of Operational Research*.
- [J.2] (JCR Q1; ABS 3) Wang, S., Yao, S., Zhang, H., Liu, Q., & Wei, L. (2025). Arc-flow formulation and branch-and-price-and-cut algorithm for the bin-packing problem with fragile objects. *Computers & Operations Research*.
- [J.3] (JCR Q1; ABS 3) Wang, S., Wang, R., Zhang, H., Jing, F., Liu, Q., & Wei, L. (2025). A goal-driven iterated local search approach based on the maximal-space for the circle bin-packing problem with rectangular items. *Computers & Operations Research*.
- [J.4] (JCR Q2; CCF A) Wang, S., Qiu, Q., Zhang, H., Wei, L., & Liu, Q. (2025). An efficient algorithm for exact SRAM verification via novel pattern matching techniques. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*.
- [J.5] (JCR Q1; ABS 3) Yao, S., Wang, S., Zhang, T., Wei, L., & Liu, Q. (2025). A combinatorial Benders decomposition approach for the constrained two-dimensional guillotine cutting problem with defects. *International Journal of Production Research*.

CONFERENCE

- [C.1] (CORE C) Wang, S., Baldacci, R., Furini, F., Wei, L., & Liu, Q. (2025). Highly effective exact algorithms for solving two-dimensional packing problems. *The 38th Annual Conference of the European Chapter on Combinatorial Optimization*.

IN-REVIEW PAPERS

- [R.1] (JCR Q2; UTD24) Wang, S., Baldacci, R., Furini, F., Wei, L., & Liu, Q. (2025). Exact algorithms for two-dimensional knapsack problems: A unified framework with new benchmark results. (Under Minor Revision) *INFORMS Journal on Computing*.
- [R.2] (JCR Q2; UTD24; FT50; ABS 4*) Wang, S., Baldacci, R., Furini, F., Wei, L., & Liu, Q. (2025). A highly efficient numerically exact algorithm for two-dimensional bin-packing problems. (Under Review) *Operations Research*.

SELECTED WORKING PAPERS

- [P.1] (JCR Q1; ABS 3) Wang, S., You, Z., Baldacci, R., Wei, L., & Xu, Z. A unified exact algorithm with novel bounding techniques for solving vehicle routing problems with two-dimensional loading constraints. Targeted at *Transportation Science*.
- [P.2] (JCR Q1; ABS 4) Wang, S., Baldacci, R., Furini, F., Wei, L., & Xu, Z. A numerically exact branch-price-and-cut algorithm for two-dimensional bin-packing problems with variable bin sizes and costs. Targeted at *European Journal of Operational Research*.
- [P.3] (JCR Q1; ABS 3) Zhang, W., Wang, S. (Co-First), Wei, L., & Liu, Q. A new branch-and-price-based heuristic algorithm for the three-dimensional bin-packing problem with stacking constraints. Targeted at *International Journal of Production Research*.
- [P.4] (JCR Q2; CCF A) Chen, Z., Wang, S. (Co-First), Wei, L., & Liu, Q. (2025). Learning-driven skyline-based variable neighborhood search for fast floorplanning. Targeted at *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*.

IV. SELECTED RESEARCH EXPERIENCE

- **New Exact Techniques for the Two-Dimensional Knapsack Problem with Guillotine Constraints** October 2024 - June 2025
(Principal Investigator)
This research produced 1 publication [J.1]
 - An improved bidirectional tree search approach and a novel iterative combination enumeration are proposed.
 - An adapted preprocessing procedure and two enhanced upper bounds are introduced.
 - 46 better solutions and 109 tighter upper bounds for many instances are computed.
 - A comprehensive open-source repository, including algorithm code and detailed results, is provided.
- **An Exact Algorithm based on Novel Pattern Matching Techniques for SRAM Verification** October 2023 - August 2025
(Principal Investigator)
This research produced 2 competition awards, 1 patent, and 1 publication [J.4]
 - The proposed SRAM verification algorithm incorporates three key innovations.
 - Compared to SOTA techniques, our algorithm achieves speedups ranging from 2× to 3789× while maintaining 100% accuracy.
- **New Model and Exact Algorithm for the Bin-Packing Problem with Fragile Objects** May 2023 - October 2024
(Principal Investigator)
This research produced 1 publication [J.2]
 - The first arc-flow formulation and branch-price-and-cut algorithm for solving the problem are proposed.
 - Compared to the SOTA approach, an additional 88 optimal solutions were obtained with 55% less computation time.

-----ONGOING RESEARCH-----

- **A Learning-Driven Skyline-Based Meta-Heuristic Algorithm for Fast Floorplanning** October 2022 - Present
(Principal Investigator)
This research produced 1 competition award, 1 patent, and 1 publication [P.4]
 - Binary classification and graph convolutional neural network are used to accelerate the proposed meta-heuristic.
- **New (Numerically) Exact and Heuristic Approaches for a Class of Two-Dimensional Packing Problems** April 2022 - Present
(Principal Investigator)
This research produced 2 patents and 5 publications [J.1], [J.3], [J.5], [C.1], [R.2], and [R.1], and is expected to produce [P.2]
 - Our approaches were introduced at a famous international conference and received positive comments from peers [C.1].
 - Our methods outperform SOTA algorithms in seven problems and can be adapted to solve other variants ([P.1], [P.2], and [P.3]).
 - A numerically exact algorithm is proposed for the two-dimensional bin-packing problem with and without guillotine cuts [R.2].
 - The algorithm integrates efficient pricing algorithms ([J.1], [J.5], and [R.1]), valid cuts [J.2], and novel primal heuristics [J.3].

V. ENTERPRISE PROJECTS

- **Cutting Optimization of Irregular Materials** January 2023 - December 2023
(Main Participant)
This project was in collaboration with Sany Heavy Industry Co., Ltd
 - My contributions: DXF file parsing, SVG conversion, graphic discretization, and lead-out line generation.
- **Production Optimization of Brightness-Enhancement Film Semi-Finished Products** June 2021 - October 2023
(Algorithm Leader)
This project was in collaboration with 3M Company and produced 2 patents (including a US patent)
 - My contributions: implementation of all algorithm components and part of the system development.
- **Cutting Optimization of Steel Pipes** November 2021 - June 2023
(Algorithm Leader)
This project was in collaboration with Bohai Shipbuilding Heavy Industry Co., Ltd and produced 1 patent
 - My contributions: implementation of all algorithm components and part of the system development.
- **Monitoring of Fan Rotor Assembly Process** December 2021 - April 2023
(Algorithm Leader)
This project was in collaboration with China Aero Engine Corporation and produced 1 patent
 - My contributions: implementation of all algorithm components and part of the system development.
- **Scheduling Optimization of Commuter Bus** October 2022 - December 2022
(Algorithm Leader)
This project was in collaboration with Lanzhou University of Information Science and Technology
 - My contributions: implementation of all algorithm components.

VI. ADDITIONAL INFORMATION

Specialized Area: Cutting and Packing;

Programming Skills: Java; Python; C/C++; Matlab;

Languages & Tests: Mandarin (native); English (IELTS: Overall 6.5; All bands ≥ 5.5); GMAT: 655

Interests: Operations Research; Combinatorial Optimization; Exact Solutions; Programming;

Major Courses: Operations Research (95/100); Numerical Analysis (93/100); Mathematics Modeling (92/100);

Research Skills:

- **Basic Exact Methods:** (Integer) Linear Programming, Dynamic Programming, Branch-and-Bound ...
- **Advanced Exact Methods:** Branch(-Cut)-and-Price, (Logic-Based) Benders Decomposition ...
- **Meta-Heuristics:** Tabu Search, Variable Neighborhood Search, Iterated Local Search ...
- **Swarm Intelligence Heuristics:** Particle Swarm, Artificial Fish Swarm, Artificial Bee Swarm ...
- **Classic Machine Learning:** KNN, SVM, Tree-Based Methods (Random Forest, XGBoost) ...
- **Deep (Reinforcement) Learning:** MLP, CNN (Resnet), RNN (LSTM), GNN (GCN), DQN, PPO, AC, DDPG ...
- **Data Analysis:** Grey Relation Analysis, Fuzzy Comprehensive Evaluation ...

VII. REFERENCES

1. **Lijun Wei** 
Full Professor, School of Electromechanical Engineering
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Relationship: Chief-Supervisor
2. **Roberto Baldacci** 
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Relationship: Co-Supervisor and Co-Author of Significant Paper
3. **Fabio Furini** 
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